THE COMPOSITION ARROW MAPS THE REDUCED CLOSURE OF THE IMAGE INTO ITSELF SET THEORETICALLY

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Allowed standard prerequisites (editorial).

- The morphism of groupoids over a field situation fixing R_1, R_2, a and the structure maps (more-groupoids chapter, Situation situation-morphism-groupoids-on-field, canonical tag 04Q5; kind reference). cited at statement lines 1754-1755. Supplies the notation and assumptions of the statement.
- Definition of the reduced induced scheme structure on a closed subset (Schemes chapter, Definition schemes-definition-reduced-induced-scheme, canonical tag 01J4). cited at statement lines 1756-1757. Fixes Z as the reduced closed subscheme of R_2 supported on the closure of the image of a.
- The closure of the image of a product map is the product of the closures of the images, set theoretically (Varieties chapter, Lemma varieties-lemma-closure-image-product-map, canonical tag 04Q0). cited at proof lines 1772-1775. Identifies the closure of the image of a x a with Z x_{s_2,U,t_2} Z, which the commutative square then maps into Z under the bottom arrow.

Source reference. Original reference 04Q5: sources/stacks-chow/tags.txt, source lines 6138–6138. Cited at statement lines 1754-1755: notation and assumptions fixing R_1, R_2, a and the structure maps.

Source reference. Original reference 01J4: sources/stacks-chow/tags.txt, source lines 2001–2001. Cited at statement lines 1756-1757: Z is the reduced closed subscheme whose underlying space is the closure of the image of a.

Source reference. Original lemma 04Q0: sources/stacks-chow/tags.txt, source lines 6133–6133. Cited at proof lines 1772-1775: the closure of the image of a x a is, set theoretically, Z x_{s_2,U,t_2} Z.

Lemma 0.2. *Notation and assumptions as in Situation 04Q5. Let $Z \subset R_2$ be the reduced closed subscheme (see Schemes, Definition 01J4) whose underlying topological space is the closure of the image of $a : R_1 \to R_2$. Then $c_2(Z \times_{s_2,U,t_2} Z) \subset Z$ set theoretically.*

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Proof. Consider the commutative diagram

$$\begin{array}{ccc} R_1 \times_{s_1,U,t_1} R_1 & \longrightarrow & R_1 \\ \downarrow & & \downarrow \\ R_2 \times_{s_2,U,t_2} R_2 & \longrightarrow & R_2 \end{array}$$

By Varieties, Lemma 04Q0 the closure of the image of the left vertical arrow is (set theoretically) $Z \times_{s_2,U,t_2} Z$. Hence the result follows. □